aj07154_HW2

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1 HW 2: Data Wrangling

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1.0.1 Q1

[4]:
```
import pandas as pd
import numpy as np #Importing relavent libraries
DataF = pd.read_csv('580SurveyCleanup.csv') #Importing .csv file and creating a␣
↪DataFrame Data Structure.
DataF.head() #View top entries to get general idea of the data conditions.
```

[4]:
```
  Unnamed: 0       Unnamed: 1 Unnamed: 2          Unnamed: 3   Unnamed: 4  \
0        No.             Time     Status  Duration (Seconds)  Access Code
1          1  10/3/20 5:49 PM  Completed               11646          NaN
2          2  10/3/20 5:49 PM  Completed                 113          NaN
3          3  10/3/20 5:49 PM  Completed                 572          NaN
4          4  10/3/20 5:50 PM  Completed                 365          NaN

      Unnamed: 5      Unnamed: 6     Unnamed: 7 Unnamed: 8  \
0  Email Address      IP Address        Country     Region
1            NaN  100.36.167.226  United States   Virginia
2            NaN   98.169.26.134  United States   Virginia
3            NaN    98.169.52.23  United States   Virginia
4            NaN  69.255.182.253  United States   Virginia

                Unnamed: 9  …            Q11.1         Q11.2  \
0  City / Town / District   …  Some familiarity  Average user
1                  Fairfax  …               NaN           NaN
2                  Fairfax  …               NaN             1
3          Greenway Downs   …               NaN           NaN
4              Gainesville  …               NaN           NaN

                      Q11.3          Q11.4         Q12            Q12.1  \
0  Frequent use for projects  Fluent/expert  Little/none  Some familiarity
1                        NaN            NaN          NaN                 1
2                        NaN            NaN            1               NaN
3                        NaN            NaN          NaN               NaN
```

```
4                          1            NaN          NaN                  1

         Q12.2                      Q12.3          Q12.4  \
0  Average user  Frequent use for projects  Fluent/expert
1           NaN                        NaN            NaN
2           NaN                        NaN            NaN
3             1                        NaN            NaN
4           NaN                        NaN            NaN

                                              Q13
0  What is your goal for learning about data anal…
1                                             NaN
2                                             NaN
3  As part of my work, I have access to a lot of …
4                                             NaN

[5 rows x 46 columns]
```

[5]:
```
DataF.columns = DataF.iloc[0,:] #Setting First Row as Column Names.
DataF = DataF[1:] #Removing First Row
DataF.head()
```

[5]:
```
0 No.             Time     Status Duration (Seconds) Access Code  \
1   1  10/3/20 5:49 PM  Completed              11646         NaN
2   2  10/3/20 5:49 PM  Completed                113         NaN
3   3  10/3/20 5:49 PM  Completed                572         NaN
4   4  10/3/20 5:50 PM  Completed                365         NaN
5   5  10/3/20 5:51 PM  Completed                 92         NaN

0 Email Address      IP Address        Country    Region  \
1           NaN  100.36.167.226  United States  Virginia
2           NaN   98.169.26.134  United States  Virginia
3           NaN    98.169.52.23  United States  Virginia
4           NaN  69.255.182.253  United States  Virginia
5           NaN  138.88.108.230  United States       NaN

0 City / Town / District  … Some familiarity Average user  \
1                Fairfax  …              NaN          NaN
2                Fairfax  …              NaN            1
3         Greenway Downs  …              NaN          NaN
4            Gainesville  …              NaN          NaN
5                    NaN  …                1          NaN

0 Frequent use for projects Fluent/expert Little/none Some familiarity  \
1                       NaN           NaN         NaN                1
2                       NaN           NaN           1              NaN
3                       NaN           NaN         NaN              NaN
```

```
4                        1            NaN          NaN                1
5                      NaN            NaN          NaN              NaN

0 Average user Frequent use for projects Fluent/expert  \
1          NaN                      NaN           NaN
2          NaN                      NaN           NaN
3            1                      NaN           NaN
4          NaN                      NaN           NaN
5            1                      NaN           NaN

0 What is your goal for learning about data analytics?
1                                                NaN
2                                                NaN
3  As part of my work, I have access to a lot of …
4                                                NaN
5                                              Learn

[5 rows x 46 columns]
```

[6]:
```
DataF.reset_index(drop=True, inplace=True) #Reseting INdex
DataF = DataF.iloc[:,1:12] #Select All  Rows, Ignore zeroth column (which simply was for indexing purposes in data and is getting redundant) upto 12th (Not including questions)
DataF.head()
```

[6]:
```
0             Time     Status Duration (Seconds) Access Code Email Address  \
0  10/3/20 5:49 PM  Completed              11646         NaN           NaN
1  10/3/20 5:49 PM  Completed                113         NaN           NaN
2  10/3/20 5:49 PM  Completed                572         NaN           NaN
3  10/3/20 5:50 PM  Completed                365         NaN           NaN
4  10/3/20 5:51 PM  Completed                 92         NaN           NaN

0      IP Address        Country    Region City / Town / District  Latitude  \
0  100.36.167.226  United States  Virginia                Fairfax  -77.2891
1   98.169.26.134  United States  Virginia                Fairfax   -77.263
2    98.169.52.23  United States  Virginia         Greenway Downs  -77.2331
3  69.255.182.253  United States  Virginia            Gainesville  -77.6308
4  138.88.108.230  United States       NaN                    NaN   -97.822

0 Longitude
0   38.8209
1   38.8605
2   38.9645
3   38.8175
4    37.751
```

Just realized that columns 12 onwards had their titles in different row. So importing .csv file into another dataframe variable.

[7]:
```python
Questions = pd.read_csv('580SurveyCleanup.csv')
Questions = Questions.iloc[:,[12,13]] #Selecting Question 1
Questions.head()
```

[7]:
```
    Q1  Q1.1
0  2.0   4.0
1  NaN   1.0
2  1.0   NaN
3  1.0   NaN
4  1.0   NaN
```

[8]:
```python
Questions.columns = Questions.iloc[0].astype(str)
Questions = Questions[1:]
Questions.head()
```

[8]:
```
0  2.0  4.0
1  NaN  1.0
2  1.0  NaN
3  1.0  NaN
4  1.0  NaN
5  1.0  NaN
```

[10]:
```python
def Populate(DataFramee,column_lst):
    DataFramee = DataFramee.fillna('') #Removing NAN Values
    for column in column_lst:
        DataFramee[column] = DataFramee[column].replace(1.0, column) #Replacing␣
 ↪1.0 with column name
    return DataFramee
def Merge(DataFramee, column_lst, New_Name):
    DataFramee[New_Name] = '' #Initializing a new column
    for column in column_lst:
        DataFramee[New_Name] += DataFramee[column] #Adding Columns
    DataFramee.drop(columns=column_lst, inplace=True) #Dropping Old Columns
    return DataFramee
Questions=Populate(Questions,['2.0','4.0'])
Questions #Test
```

[10]:
```
0   2.0  4.0
1        4.0
2   2.0
3   2.0
4   2.0
5   2.0
6   2.0
7   2.0
8   2.0
```

```
9   2.0
10       4.0
11  2.0
12       4.0
13  2.0
14  2.0
15       4.0
16       4.0
17  2.0
18  2.0
19       4.0
20       4.0
21       4.0
22  2.0
23       4.0
24       4.0
25       4.0
26  2.0
27  2.0
28       4.0
29       4.0
30  2.0
31       4.0
32  2.0
33  2.0
34       4.0
35  2.0
36       4.0
```

[11]:
```
Questions=Merge(Questions,['2.0','4.0'],'Question 1')
Questions.head
```

[11]:
```
<bound method NDFrame.head of 0  Question 1
1          4.0
2          2.0
3          2.0
4          2.0
5          2.0
6          2.0
7          2.0
8          2.0
9          2.0
10         4.0
11         2.0
12         4.0
13         2.0
14         2.0
```

```
15          4.0
16          4.0
17          2.0
18          2.0
19          4.0
20          4.0
21          4.0
22          2.0
23          4.0
24          4.0
25          4.0
26          2.0
27          2.0
28          4.0
29          4.0
30          2.0
31          4.0
32          2.0
33          2.0
34          4.0
35          2.0
36          4.0>
```

[12]:
```python
Questions.reset_index(drop=True, inplace=True) #Reseting Index
Questions
```

[12]:
```
0  Question 1
0           4.0
1           2.0
2           2.0
3           2.0
4           2.0
5           2.0
6           2.0
7           2.0
8           2.0
9           4.0
10          2.0
11          4.0
12          2.0
13          2.0
14          4.0
15          4.0
16          2.0
17          2.0
18          4.0
19          4.0
```

```
20         4.0
21         2.0
22         4.0
23         4.0
24         4.0
25         2.0
26         2.0
27         4.0
28         4.0
29         2.0
30         4.0
31         2.0
32         2.0
33         4.0
34         2.0
35         4.0
```

Looks like we will have to tailor every question individually and then merge everything in the end.

[17]:
```
Questions2 = pd.read_csv('580SurveyCleanup.csv')
Questions2 = Questions2.iloc[:, [14, 15, 16, 17]]
Questions2.columns = Questions2.iloc[0].astype(str)
Questions2 = Questions2[1:]
Questions2
```

[17]:

0	Male	Female	NonBinary	Prefer not to answer
1	NaN	1	NaN	NaN
2	NaN	1	NaN	NaN
3	1	NaN	NaN	NaN
4	NaN	NaN	NaN	1
5	1	NaN	NaN	NaN
6	NaN	1	NaN	NaN
7	NaN	NaN	1	NaN
8	NaN	1	NaN	NaN
9	1	NaN	NaN	NaN
10	1	NaN	NaN	NaN
11	NaN	1	NaN	NaN
12	NaN	1	NaN	NaN
13	1	NaN	NaN	NaN
14	NaN	1	NaN	NaN
15	NaN	1	NaN	NaN
16	1	NaN	NaN	NaN
17	1	NaN	NaN	NaN
18	1	NaN	NaN	NaN
19	1	NaN	NaN	NaN
20	NaN	1	NaN	NaN
21	1	NaN	NaN	NaN

```
22  NaN      1       NaN                    NaN
23  NaN      1       NaN                    NaN
24    1    NaN       NaN                    NaN
25  NaN      1       NaN                    NaN
26    1    NaN       NaN                    NaN
27    1    NaN       NaN                    NaN
28    1    NaN       NaN                    NaN
29  NaN      1       NaN                    NaN
30    1    NaN       NaN                    NaN
31  NaN      1       NaN                    NaN
32  NaN      1       NaN                    NaN
33    1    NaN       NaN                    NaN
34    1    NaN       NaN                    NaN
35    1    NaN       NaN                    NaN
36  NaN      1       NaN                    NaN
```

[18]:
```
def Populate_v2(DataFramee,column_lst,replacement_value): #Modifying Populate
    DataFramee = DataFramee.fillna('') #Removing NAN Values
    for column in column_lst:
        DataFramee[column] = DataFramee[column].replace(replacement_value,␣
 ↪column) #Replacing 1 with column name
    return DataFramee

Questions2=Populate_v2(Questions2, ['Male', 'Female', 'NonBinary', 'Prefer not␣
 ↪to answer'],'1') #Passing a list manually containing column names
Questions2
```

[18]:
```
0   Male  Female  NonBinary  Prefer not to answer
1         Female
2         Female
3   Male
4                            Prefer not to answer
5   Male
6         Female
7                 NonBinary
8         Female
9   Male
10  Male
11        Female
12        Female
13  Male
14        Female
15        Female
16  Male
17  Male
18  Male
19  Male
```

```
20        Female
21  Male
22        Female
23        Female
24  Male
25        Female
26  Male
27  Male
28  Male
29        Female
30  Male
31        Female
32        Female
33  Male
34  Male
35  Male
36        Female
```

[19]:
```
Questions2 = Merge(Questions2,['Male', 'Female', 'NonBinary', 'Prefer not to␣
↪answer'],'Question 2')
Questions2.reset_index(drop=True, inplace=True) #Reseting Index
Questions2
```

[19]:
```
0            Question 2
0                Female
1                Female
2                  Male
3   Prefer not to answer
4                  Male
5                Female
6             NonBinary
7                Female
8                  Male
9                  Male
10               Female
11               Female
12                 Male
13               Female
14               Female
15                 Male
16                 Male
17                 Male
18                 Male
19               Female
20                 Male
21               Female
22               Female
```

```
23                    Male
24                  Female
25                    Male
26                    Male
27                    Male
28                  Female
29                    Male
30                  Female
31                  Female
32                    Male
33                    Male
34                    Male
35                  Female
```

Testing Merge Function on Question 8 of Survey

[26]:
```
Questions8 = pd.read_csv('580SurveyCleanup.csv')
Questions8 = Questions8.iloc[:,[23,24,25,26]]
Questions8.columns = Questions8.iloc[0].astype(str)
Questions8 = Questions8[1:]
Questions8
```

[26]:
```
0  Microsoft/Windows Apple/MacBook Other           nan
1                NaN             1   NaN           NaN
2                NaN             1   NaN           NaN
3                NaN             1   NaN           NaN
4                  1           NaN   NaN           NaN
5                NaN             2   NaN           NaN
6                NaN           NaN   NaN           NaN
7                  1           NaN   NaN           NaN
8                NaN           NaN   NaN           NaN
9                  1           NaN   NaN           NaN
10               NaN           NaN   NaN           NaN
11                 1           NaN   NaN           NaN
12                 1           NaN   NaN           NaN
13               NaN             1   NaN           NaN
14                 1           NaN   NaN           NaN
15                 1           NaN   NaN           NaN
16                 1           NaN   NaN           NaN
17                 1             1   NaN           NaN
18                 1           NaN   NaN           NaN
19                 1           NaN   NaN           NaN
20               NaN             1   NaN           NaN
21                 1           NaN   NaN           NaN
22                 1           NaN   NaN           NaN
23                 1             1   NaN           NaN
24                 1           NaN   NaN           NaN
25                 1           NaN   NaN           NaN
```

```
26                0          NaN   NaN           NaN
27              NaN            1   NaN           NaN
28                1          NaN   NaN           NaN
29                1          NaN   NaN           NaN
30              NaN            1   NaN           NaN
31                1          NaN   NaN           NaN
32                1            1   NaN           NaN
33                1          NaN   NaN           NaN
34                1          NaN   NaN           NaN
35                1          NaN   NaN           NaN
36              NaN          NaN     1  Acer/Windows
```

[27]:
```
Questions8=Populate_v2(Questions8,['Microsoft/Windows', 'Apple/
↪MacBook','Other','nan'],['0','1','2'])
```

[28]:
```
Questions8=Merge(Questions8,['Microsoft/Windows', 'Apple/
↪MacBook','Other','nan'],'Question 8')
Questions8.reset_index(drop=True, inplace=True) #Reseting Index
Questions8
```

[28]:
```
0                        Question 8
0                     Apple/MacBook
1                     Apple/MacBook
2                     Apple/MacBook
3                 Microsoft/Windows
4                     Apple/MacBook
5
6                 Microsoft/Windows
7
8                 Microsoft/Windows
9
10                Microsoft/Windows
11                Microsoft/Windows
12                    Apple/MacBook
13                Microsoft/Windows
14                Microsoft/Windows
15                Microsoft/Windows
16  Microsoft/WindowsApple/MacBook
17                Microsoft/Windows
18                Microsoft/Windows
19                    Apple/MacBook
20                Microsoft/Windows
21                Microsoft/Windows
22  Microsoft/WindowsApple/MacBook
23                Microsoft/Windows
24                Microsoft/Windows
25                Microsoft/Windows
```

```
26                     Apple/MacBook
27                 Microsoft/Windows
28                 Microsoft/Windows
29                     Apple/MacBook
30                 Microsoft/Windows
31    Microsoft/WindowsApple/MacBook
32                 Microsoft/Windows
33                 Microsoft/Windows
34                 Microsoft/Windows
35                 OtherAcer/Windows
```

Using Same Method for Question 9

[30]:
```
Questions9 = pd.read_csv('580SurveyCleanup.csv')
Questions9 = Questions9.iloc[:,[27,28,29]]
Questions9.columns = Questions9.iloc[0].astype(str)
Questions9 = Questions9[1:]
Questions9=Populate_v2(Questions9,['Yes, Full Time', 'Working, but not Full␣
↪Time','Not Working while attending Mason'],'1')
Questions9=Merge(Questions9,['Yes, Full Time', 'Working, but not Full␣
↪Time','Not Working while attending Mason'],'Question 9')
Questions9.reset_index(drop=True, inplace=True) #Reseting Index
Questions9
```

[30]:
```
0                          Question 9
0   Not Working while attending Mason
1              Working, but not Full Time
2                          Yes, Full Time
3   Not Working while attending Mason
4                          Yes, Full Time
5

6              Working, but not Full Time
7

8                          Yes, Full Time
9

10  Not Working while attending Mason
11  Not Working while attending Mason
12             Working, but not Full Time
13  Not Working while attending Mason
14  Not Working while attending Mason
15  Not Working while attending Mason
16                         Yes, Full Time
17  Not Working while attending Mason
18  Not Working while attending Mason
19             Working, but not Full Time
20                         Yes, Full Time
21  Not Working while attending Mason
22             Working, but not Full Time
```

```
23  Not Working while attending Mason
24                       Yes, Full Time
25                       Yes, Full Time
26  Not Working while attending Mason
27  Not Working while attending Mason
28  Not Working while attending Mason
29  Not Working while attending Mason
30                       Yes, Full Time
31  Not Working while attending Mason
32                       Yes, Full Time
33                       Yes, Full Time
34  Not Working while attending Mason
35                       Yes, Full Time
```

Questions 10-12 are exactly same, so copy pasting the same code but creating different dataframes.

[31]:
```
Questions10 = pd.read_csv('580SurveyCleanup.csv')
Questions10 = Questions10.iloc[:,[30,31,32,33,34]]
Questions10.columns = Questions10.iloc[0].astype(str)
Questions10 = Questions10[1:]

Questions10=Populate_v2(Questions10,['Little/none', 'Some familiarity','Average␣
↪user','Frequent use for projects' , 'Fluent/expert'],'1')
Questions10=Merge(Questions10,['Little/none','Some familiarity','Average␣
↪user','Frequent use for projects','Fluent/expert'],'Question 10')
Questions10.reset_index(drop=True, inplace=True) #Reseting Index
Questions10
```

[31]:
```
0                Question 10
0           Some familiarity
1               Average user
2           Some familiarity
3               Average user
4           Some familiarity
5
6               Average user
7
8           Some familiarity
9
10               Little/none
11  Frequent use for projects
12          Some familiarity
13              Average user
14              Average user
15          Some familiarity
16              Average user
17  Frequent use for projects
```

```
18                Little/none
19           Some familiarity
20           Some familiarity
21           Some familiarity
22           Some familiarity
23                Little/none
24               Average user
25               Average user
26               Average user
27               Average user
28               Average user
29               Average user
30                Little/none
31               Average user
32               Average user
33               Average user
34           Some familiarity
35           Some familiarity
```

[32]:

```python
Questions11 = pd.read_csv('580SurveyCleanup.csv')
Questions11 = Questions11.iloc[:,[35,36,37,38,39]]
Questions11.columns = Questions11.iloc[0].astype(str)
Questions11 = Questions11[1:]

Questions11=Populate_v2(Questions11,['Little/none', 'Some familiarity','Average␣
 ↪user','Frequent use for projects' , 'Fluent/expert'],'1')
Questions11=Merge(Questions11,['Little/none','Some familiarity','Average␣
 ↪user','Frequent use for projects','Fluent/expert'],'Question 11')
Questions11.reset_index(drop=True, inplace=True) #Reseting Index
Questions11
```

[32]:

```
0                 Question 11
0                 Little/none
1                Average user
2                 Little/none
3   Frequent use for projects
4            Some familiarity
5
6                Average user
7
8   Frequent use for projects
9
10                Little/none
11               Average user
12               Average user
13               Average user
14                Little/none
```

```
15           Some familiarity
16           Some familiarity
17  Frequent use for projects
18                Little/none
19             Fluent/expert
20           Some familiarity
21                Little/none
22  Frequent use for projects
23                Little/none
24               Average user
25           Some familiarity
26  Frequent use for projects
27  Frequent use for projects
28               Average user
29               Average user
30           Some familiarity
31  Frequent use for projects
32           Some familiarity
33               Average user
34           Some familiarity
35                Little/none
```

[60]:

```
Data = pd.read_csv('580SurveyCleanup.csv')
Questions12 = Data.iloc[:,[40,41,42,43,44]]
Questions13 = Data.iloc[:,[45]]
Questions13 = Questions13[1:].fillna('')
Questions13.columns=['Question 13']
Questions12.columns = Questions12.iloc[0].astype(str)
Questions12 = Questions12[1:]

Questions12=Populate_v2(Questions12,['Little/none', 'Some familiarity','Average␣
↪user','Frequent use for projects' , 'Fluent/expert'],'1')
Questions12=Merge(Questions12,['Little/none','Some familiarity','Average␣
↪user','Frequent use for projects','Fluent/expert'],'Question 12')
Questions12.reset_index(drop=True, inplace=True) #Reseting Index
Questions13.reset_index(drop=True, inplace=True) #Reseting Index
Questions8to13=pd.
↪concat([Questions8,Questions9,Questions10,Questions11,Questions12,Questions13],axis=1)
Questions8to13
```

[60]:

```
                 Question 8                       Question 9  \
0             Apple/MacBook  Not Working while attending Mason
1             Apple/MacBook         Working, but not Full Time
2             Apple/MacBook                    Yes, Full Time
3         Microsoft/Windows  Not Working while attending Mason
4             Apple/MacBook                    Yes, Full Time
5
```

```
6                Microsoft/Windows        Working, but not Full Time
7
8                Microsoft/Windows                     Yes, Full Time
9
10               Microsoft/Windows  Not Working while attending Mason
11               Microsoft/Windows  Not Working while attending Mason
12                   Apple/MacBook        Working, but not Full Time
13               Microsoft/Windows  Not Working while attending Mason
14               Microsoft/Windows  Not Working while attending Mason
15               Microsoft/Windows  Not Working while attending Mason
16  Microsoft/WindowsApple/MacBook                     Yes, Full Time
17               Microsoft/Windows  Not Working while attending Mason
18               Microsoft/Windows  Not Working while attending Mason
19                   Apple/MacBook        Working, but not Full Time
20               Microsoft/Windows                     Yes, Full Time
21               Microsoft/Windows  Not Working while attending Mason
22  Microsoft/WindowsApple/MacBook        Working, but not Full Time
23               Microsoft/Windows  Not Working while attending Mason
24               Microsoft/Windows                     Yes, Full Time
25               Microsoft/Windows                     Yes, Full Time
26                   Apple/MacBook  Not Working while attending Mason
27               Microsoft/Windows  Not Working while attending Mason
28               Microsoft/Windows  Not Working while attending Mason
29                   Apple/MacBook  Not Working while attending Mason
30               Microsoft/Windows                     Yes, Full Time
31  Microsoft/WindowsApple/MacBook  Not Working while attending Mason
32               Microsoft/Windows                     Yes, Full Time
33               Microsoft/Windows                     Yes, Full Time
34               Microsoft/Windows  Not Working while attending Mason
35               OtherAcer/Windows                     Yes, Full Time

                Question 10                Question 11  \
0          Some familiarity                Little/none
1              Average user               Average user
2          Some familiarity                Little/none
3              Average user  Frequent use for projects
4          Some familiarity           Some familiarity
5
6              Average user               Average user
7
8          Some familiarity  Frequent use for projects
9
10              Little/none                Little/none
11  Frequent use for projects             Average user
12         Some familiarity               Average user
13             Average user               Average user
14             Average user                Little/none
```

```
15          Some familiarity          Some familiarity
16              Average user          Some familiarity
17  Frequent use for projects  Frequent use for projects
18               Little/none               Little/none
19          Some familiarity             Fluent/expert
20          Some familiarity          Some familiarity
21          Some familiarity               Little/none
22          Some familiarity  Frequent use for projects
23               Little/none               Little/none
24              Average user              Average user
25              Average user          Some familiarity
26              Average user  Frequent use for projects
27              Average user  Frequent use for projects
28              Average user              Average user
29              Average user              Average user
30               Little/none          Some familiarity
31              Average user  Frequent use for projects
32              Average user          Some familiarity
33              Average user              Average user
34          Some familiarity          Some familiarity
35          Some familiarity               Little/none

                 Question 12  \
0           Some familiarity
1                Little/none
2               Average user
3           Some familiarity
4               Average user
5
6               Average user
7
8               Average user
9
10          Some familiarity
11  Frequent use for projects
12              Average user
13              Average user
14               Little/none
15          Some familiarity
16          Some familiarity
17              Average user
18              Average user
19              Average user
20          Some familiarity
21              Average user
22               Little/none
23               Little/none
```

24	Average user
25	Some familiarity
26	Average user
27	Average user
28	Average user
29	Frequent use for projects
30	Some familiarity
31	Frequent use for projects
32	Some familiarity
33	Average user
34	Some familiarity
35	Little/none

	Question 13
0	
1	
2	As part of my work, I have access to a lot of …
3	
4	Learn
5	
6	To deduce meaningful interpretation for Financ…
7	
8	
9	
10	To see if it is a field I am interested in pur…
11	To learn and explore cutting edge data technol…
12	Learn to select the appropriate tools to condu…
13	
14	improve my skills
15	My professional goal is to become a Financial …
16	able to compliment my Computational Social Sci…
17	To start a company
18	To be able understand and manipulate data to m…
19	My ultimate goal is to get a job as a sports a…
20	To understand how to properly interpret and di…
21	After graduating data analytics, I want myself…
22	Learning how to appropriately handle large qua…
23	To learn and understand the fundamentals and m…
24	The goal is to learn as many programming/stati…
25	Learning analytics tool and being good at them
26	I want to be as a data analyst
27	It is very relevant to my PhD research area an…
28	Learn how to handle large amount structured an…
29	To become a good Data Analyst and work for top…
30	Get a better understanding of the granularity …
31	
32	Learn how data analytics works

```
33                     Advance my career in data science
34                                    Getting a good job
35                              to use the skills at work.
```

Merging DataFrames

[61]:
```
Questions3to7 = Data.iloc[:,[18,19,20,21,22]]
Questions3to7 = Questions3to7[1:]
NewColumnNames=['Question 3','Question 4','Question 5','Question 6','Question␣
 ↪7'] #Mistakenly Renamed them differently before, so doing thiss again for␣
 ↪consistency
Questions3to7.columns=NewColumnNames
Questions3to7.reset_index(drop=True, inplace=True) #Reseting Index
FinalDataFrame=pd.
 ↪concat([DataF,Questions,Questions2,Questions3to7,Questions8to13],axis=1)
FinalDataFrame.to_csv('formatted_580SurveyCleanup.csv', index=False) #Exporting␣
 ↪Files
FinalDataFrame
```

[61]:
```
                 Time      Status Duration (Seconds) Access Code Email Address  \
0     10/3/20 5:49 PM   Completed              11646         NaN           NaN
1     10/3/20 5:49 PM   Completed                113         NaN           NaN
2     10/3/20 5:49 PM   Completed                572         NaN           NaN
3     10/3/20 5:50 PM   Completed                365         NaN           NaN
4     10/3/20 5:51 PM   Completed                 92         NaN           NaN
5     10/3/20 5:59 PM  Incomplete                221         NaN           NaN
6     10/3/20 6:23 PM   Completed             115983         NaN           NaN
7     10/3/20 7:10 PM  Incomplete                 40         NaN           NaN
8     10/3/20 7:25 PM   Completed                137         NaN           NaN
9     10/3/20 7:56 PM  Incomplete                179         NaN           NaN
10    10/3/20 8:14 PM   Completed                127         NaN           NaN
11    10/3/20 9:38 PM   Completed                269         NaN           NaN
12    10/3/20 9:41 PM   Completed                117         NaN           NaN
13   10/3/20 10:49 PM   Completed               5966         NaN           NaN
14   10/3/20 11:25 PM   Completed                254         NaN           NaN
15   10/4/20 12:28 AM   Completed                526         NaN           NaN
16    10/4/20 5:22 AM   Completed                398         NaN           NaN
17    10/4/20 5:36 AM   Completed             128253         NaN           NaN
18    10/4/20 5:47 AM   Completed                239         NaN           NaN
19   10/4/20 10:32 AM   Completed                255         NaN           NaN
20   10/4/20 11:03 AM   Completed                256         NaN           NaN
21   10/4/20 12:34 PM   Completed                674         NaN           NaN
22    10/4/20 1:36 PM   Completed                130         NaN           NaN
23    10/4/20 1:50 PM   Completed                275         NaN           NaN
24    10/4/20 9:28 PM   Completed                325         NaN           NaN
25    10/5/20 9:37 AM   Completed                257         NaN           NaN
26    10/5/20 9:36 AM   Completed                657         NaN           NaN
27   10/5/20 10:36 AM   Completed               1750         NaN           NaN
```

28	10/5/20 11:02 AM	Completed	887	NaN	NaN
29	10/5/20 11:17 AM	Completed	180	NaN	NaN
30	10/5/20 12:52 PM	Completed	392	NaN	NaN
31	10/5/20 12:56 PM	Incomplete	192	NaN	NaN
32	10/5/20 1:52 PM	Completed	277	NaN	NaN
33	10/5/20 3:34 PM	Completed	119	NaN	NaN
34	10/5/20 3:41 PM	Completed	149	NaN	NaN
35	10/5/20 5:58 PM	Completed	285	NaN	NaN

	IP Address	Country	Region	City / Town / District	Latitude \
0	100.36.167.226	United States	Virginia	Fairfax	-77.2891
1	98.169.26.134	United States	Virginia	Fairfax	-77.263
2	98.169.52.23	United States	Virginia	Greenway Downs	-77.2331
3	69.255.182.253	United States	Virginia	Gainesville	-77.6308
4	138.88.108.230	United States	NaN	NaN	-97.822
5	108.56.134.80	United States	Virginia	Reston	-77.3489
6	108.48.175.220	United States	Virginia	Arlington	-77.0987
7	173.67.199.85	United States	Virginia	Ashburn	-77.539
8	74.96.114.12	United States	Virginia	Woodbridge	-77.3197
9	107.77.241.32	United States	Texas	NaN	-96.8217
10	138.88.182.245	United States	Maryland	Takoma Park	-77.004
11	72.83.246.60	United States	Virginia	Fairfax	-77.263
12	96.241.43.105	United States	Virginia	Chantilly	-77.4512
13	71.191.72.223	United States	Virginia	Burke	-77.2866
14	73.134.53.227	United States	Maryland	Hyattsville	-76.8696
15	73.86.219.52	United States	Maryland	Baltimore	-76.6355
16	69.138.215.127	United States	Virginia	Arlington	-77.113
17	72.196.231.70	United States	Virginia	Springfield	-77.2333
18	100.15.54.17	United States	Virginia	Woodbridge	-77.3197
19	184.170.83.239	United States	Virginia	Rustburg	-79.0985
20	100.36.243.102	United States	Virginia	Fairfax	-77.3375
21	108.56.134.80	United States	Virginia	Reston	-77.3489
22	73.163.163.82	United States	Maryland	Baltimore	-76.557
23	138.88.223.87	United States	NaN	NaN	-97.822
24	100.36.50.114	United States	NaN	NaN	-97.822
25	108.51.59.195	United States	Virginia	Burke	-77.2866
26	72.83.246.219	United States	Virginia	Fairfax	-77.263
27	100.36.118.186	United States	Virginia	Burke	-77.2866
28	172.58.187.98	United States	Maryland	Baltimore	-76.6949
29	72.196.230.201	United States	Virginia	Springfield	-77.2333
30	108.18.111.204	United States	Virginia	Lorton	-77.206
31	71.191.72.223	United States	Virginia	Burke	-77.2866
32	184.191.72.8	United States	Virginia	Falls Church	-77.1425
33	141.156.160.243	United States	NaN	NaN	-97.822
34	71.206.51.121	United States	Maryland	Frederick	-77.2758
35	129.174.252.234	United States	Virginia	Fairfax	-77.3375

	…	Question 4	Question 5	Question 6	\
0	…	63	Taiwan	economics	
1	…	59	India	Finance	
2	…	70	India	Computer Applications	
3	…	66	South Korea	Business Management	
4	…	70	US	Yes	
5	…	NaN	NaN	NaN	
6	…	70	pakistan	Finance	
7	…	5.2	NaN	NaN	
8	…	71	china	Economic	
9	…	71	Uzbekistan	Economics with Finance	
10	…	66	USA	BIT	
11	…	66	India	Computer Engineering	
12	…	69	Romania	Biology	
13	…	66	India	Cse	
14	…	67	China	Management	
15	…	72	Uzbekistan	Economics with Finance	
16	…	68	USA	Economics	
17	…	60	India	Engineering	
18	…	67	Ethiopia	Accounting and ISOM	
19	…	67	United States	Statistics	
20	…	74	United States	Computer Science	
21	…	65	India	Computer science	
22	…	67	USA	Mathematics & Statistics	
23	…	75	Nigeria	Economics	
24	…	67	USA	Systems Engineering	
25	…	68	Taiwan	Civil engineering	
26	…	72	India	ECE	
27	…	61	Nigeria	Quantity Surveying	
28	…	64	India	Bachelor of Engineering	
29	…	71	India	B.Tech in CS	
30	…	56	United States	BS accounting	
31	…	70	India	Bachelor of engineering	
32	…	70	United States ; Pakistan	Criminology Law and Soci	
33	…	73	USA	Economics	
34	…	68	Vietnam	Finance	
35	…	66	Uzbekistan	Psychology	

	Question 7	Question 8	\
0	2021-12-31	Apple/MacBook	
1	2021-05-01	Apple/MacBook	
2	2022-06-01	Apple/MacBook	
3	2021-12-31	Microsoft/Windows	
4	2021-01-01	Apple/MacBook	
5	NaN		
6	2022-07-15	Microsoft/Windows	
7	NaN		

```
8   2021-12-15               Microsoft/Windows
9          NaN
10  2022-05-01               Microsoft/Windows
11  2021-05-31               Microsoft/Windows
12  2021-08-26                   Apple/MacBook
13  2021-12-25               Microsoft/Windows
14  2021-12-31               Microsoft/Windows
15  2022-05-01               Microsoft/Windows
16  2022-05-31  Microsoft/WindowsApple/MacBook
17  2021-01-12               Microsoft/Windows
18  2021-12-20               Microsoft/Windows
19  2021-12-20                   Apple/MacBook
20  2022-05-01               Microsoft/Windows
21  2022-04-16               Microsoft/Windows
22  2021-12-18  Microsoft/WindowsApple/MacBook
23  2021-05-14               Microsoft/Windows
24  2022-12-19               Microsoft/Windows
25  2021-05-30               Microsoft/Windows
26  2021-12-11                   Apple/MacBook
27  2023-05-24               Microsoft/Windows
28  2021-12-30               Microsoft/Windows
29  2021-12-15                   Apple/MacBook
30  2021-05-01               Microsoft/Windows
31  2021-12-21  Microsoft/WindowsApple/MacBook
32  2021-12-18               Microsoft/Windows
33  2022-05-15               Microsoft/Windows
34  2022-12-12               Microsoft/Windows
35  2021-12-20               OtherAcer/Windows

                           Question 9                Question 10  \
0   Not Working while attending Mason           Some familiarity
1          Working, but not Full Time               Average user
2                      Yes, Full Time           Some familiarity
3   Not Working while attending Mason               Average user
4                      Yes, Full Time           Some familiarity
5
6          Working, but not Full Time               Average user
7
8                      Yes, Full Time           Some familiarity
9
10  Not Working while attending Mason                Little/none
11  Not Working while attending Mason  Frequent use for projects
12         Working, but not Full Time           Some familiarity
13  Not Working while attending Mason               Average user
14  Not Working while attending Mason               Average user
15  Not Working while attending Mason           Some familiarity
16                     Yes, Full Time               Average user
```

```
17  Not Working while attending Mason  Frequent use for projects
18  Not Working while attending Mason                Little/none
19         Working, but not Full Time           Some familiarity
20                     Yes, Full Time           Some familiarity
21  Not Working while attending Mason           Some familiarity
22         Working, but not Full Time           Some familiarity
23  Not Working while attending Mason                Little/none
24                     Yes, Full Time               Average user
25                     Yes, Full Time               Average user
26  Not Working while attending Mason               Average user
27  Not Working while attending Mason               Average user
28  Not Working while attending Mason               Average user
29  Not Working while attending Mason               Average user
30                     Yes, Full Time                Little/none
31  Not Working while attending Mason               Average user
32                     Yes, Full Time               Average user
33                     Yes, Full Time               Average user
34  Not Working while attending Mason           Some familiarity
35                     Yes, Full Time           Some familiarity

                  Question 11                Question 12  \
0                 Little/none           Some familiarity
1                Average user                Little/none
2                 Little/none               Average user
3   Frequent use for projects           Some familiarity
4            Some familiarity               Average user
5
6                Average user               Average user
7
8   Frequent use for projects               Average user
9
10                Little/none           Some familiarity
11               Average user  Frequent use for projects
12               Average user               Average user
13               Average user               Average user
14                Little/none                Little/none
15           Some familiarity           Some familiarity
16           Some familiarity           Some familiarity
17  Frequent use for projects               Average user
18                Little/none               Average user
19              Fluent/expert               Average user
20           Some familiarity           Some familiarity
21                Little/none               Average user
22  Frequent use for projects                Little/none
23                Little/none                Little/none
24               Average user               Average user
25           Some familiarity           Some familiarity
```

26	Frequent use for projects	Average user
27	Frequent use for projects	Average user
28	Average user	Average user
29	Average user	Frequent use for projects
30	Some familiarity	Some familiarity
31	Frequent use for projects	Frequent use for projects
32	Some familiarity	Some familiarity
33	Average user	Average user
34	Some familiarity	Some familiarity
35	Little/none	Little/none

	Question 13
0	
1	
2	As part of my work, I have access to a lot of …
3	
4	Learn
5	
6	To deduce meaningful interpretation for Financ…
7	
8	
9	
10	To see if it is a field I am interested in pur…
11	To learn and explore cutting edge data technol…
12	Learn to select the appropriate tools to condu…
13	
14	improve my skills
15	My professional goal is to become a Financial …
16	able to compliment my Computational Social Sci…
17	To start a company
18	To be able understand and manipulate data to m…
19	My ultimate goal is to get a job as a sports a…
20	To understand how to properly interpret and di…
21	After graduating data analytics, I want myself…
22	Learning how to appropriately handle large qua…
23	To learn and understand the fundamentals and m…
24	The goal is to learn as many programming/stati…
25	Learning analytics tool and being good at them
26	I want to be as a data analyst
27	It is very relevant to my PhD research area an…
28	Learn how to handle large amount structured an…
29	To become a good Data Analyst and work for top…
30	Get a better understanding of the granularity …
31	
32	Learn how data analytics works
33	Advance my career in data science
34	Getting a good job

```
35                          to use the skills at work.

[36 rows x 24 columns]
```

1.1 Q2

[38]:
```python
import pandas as pd
import numpy as np
DataFrame1 = pd.read_csv('formatted_580SurveyCleanup.csv')
DataFrame1.isnull().sum()
```

[38]:
```
Time                       0
Status                     0
Duration (Seconds)         0
Access Code               36
Email Address             36
IP Address                 0
Country                    0
Region                     4
City / Town / District     5
Latitude                   0
Longitude                  0
Question 1                 0
Question 2                 0
Question 3                 0
Question 4                 1
Question 5                 2
Question 6                 2
Question 7                 3
Question 8                 3
Question 9                 3
Question 10                3
Question 11                3
Question 12                3
Question 13                9
dtype: int64
```

Here we can see clearly that Access Code and and Email address is null in its entirety. Dropping it is wise as it is taking up useful space. (Assuming the original survey did not had this mandatory, we can drop it without any severe consequences).

[39]:
```python
DataFrame1 = DataFrame1.drop(['Access Code', 'Email Address'], axis=1)
DataFrame1
```

[39]:
```
                Time     Status  Duration (Seconds)      IP Address  \
0     10/3/20 5:49 PM  Completed               11646  100.36.167.226
1     10/3/20 5:49 PM  Completed                 113   98.169.26.134
2     10/3/20 5:49 PM  Completed                 572    98.169.52.23
```

3	10/3/20 5:50 PM	Completed	365	69.255.182.253
4	10/3/20 5:51 PM	Completed	92	138.88.108.230
5	10/3/20 5:59 PM	Incomplete	221	108.56.134.80
6	10/3/20 6:23 PM	Completed	115983	108.48.175.220
7	10/3/20 7:10 PM	Incomplete	40	173.67.199.85
8	10/3/20 7:25 PM	Completed	137	74.96.114.12
9	10/3/20 7:56 PM	Incomplete	179	107.77.241.32
10	10/3/20 8:14 PM	Completed	127	138.88.182.245
11	10/3/20 9:38 PM	Completed	269	72.83.246.60
12	10/3/20 9:41 PM	Completed	117	96.241.43.105
13	10/3/20 10:49 PM	Completed	5966	71.191.72.223
14	10/3/20 11:25 PM	Completed	254	73.134.53.227
15	10/4/20 12:28 AM	Completed	526	73.86.219.52
16	10/4/20 5:22 AM	Completed	398	69.138.215.127
17	10/4/20 5:36 AM	Completed	128253	72.196.231.70
18	10/4/20 5:47 AM	Completed	239	100.15.54.17
19	10/4/20 10:32 AM	Completed	255	184.170.83.239
20	10/4/20 11:03 AM	Completed	256	100.36.243.102
21	10/4/20 12:34 PM	Completed	674	108.56.134.80
22	10/4/20 1:36 PM	Completed	130	73.163.163.82
23	10/4/20 1:50 PM	Completed	275	138.88.223.87
24	10/4/20 9:28 PM	Completed	325	100.36.50.114
25	10/5/20 9:37 AM	Completed	257	108.51.59.195
26	10/5/20 9:36 AM	Completed	657	72.83.246.219
27	10/5/20 10:36 AM	Completed	1750	100.36.118.186
28	10/5/20 11:02 AM	Completed	887	172.58.187.98
29	10/5/20 11:17 AM	Completed	180	72.196.230.201
30	10/5/20 12:52 PM	Completed	392	108.18.111.204
31	10/5/20 12:56 PM	Incomplete	192	71.191.72.223
32	10/5/20 1:52 PM	Completed	277	184.191.72.8
33	10/5/20 3:34 PM	Completed	119	141.156.160.243
34	10/5/20 3:41 PM	Completed	149	71.206.51.121
35	10/5/20 5:58 PM	Completed	285	129.174.252.234

	Country	Region	City / Town / District	Latitude	Longitude \
0	United States	Virginia	Fairfax	-77.2891	38.8209
1	United States	Virginia	Fairfax	-77.2630	38.8605
2	United States	Virginia	Greenway Downs	-77.2331	38.9645
3	United States	Virginia	Gainesville	-77.6308	38.8175
4	United States	NaN	NaN	-97.8220	37.7510
5	United States	Virginia	Reston	-77.3489	38.9311
6	United States	Virginia	Arlington	-77.0987	38.8600
7	United States	Virginia	Ashburn	-77.5390	39.0180
8	United States	Virginia	Woodbridge	-77.3197	38.6780
9	United States	Texas	NaN	-96.8217	32.7787
10	United States	Maryland	Takoma Park	-77.0040	38.9827
11	United States	Virginia	Fairfax	-77.2630	38.8605

```
12  United States  Virginia            Chantilly  -77.4512    38.9036
13  United States  Virginia                Burke  -77.2866    38.7851
14  United States  Maryland          Hyattsville  -76.8696    38.9544
15  United States  Maryland            Baltimore  -76.6355    39.3555
16  United States  Virginia            Arlington  -77.1130    38.8747
17  United States  Virginia          Springfield  -77.2333    38.7438
18  United States  Virginia           Woodbridge  -77.3197    38.6780
19  United States  Virginia             Rustburg  -79.0985    37.2544
20  United States  Virginia              Fairfax  -77.3375    38.8357
21  United States  Virginia               Reston  -77.3489    38.9311
22  United States  Maryland            Baltimore  -76.5570    39.2784
23  United States       NaN                  NaN  -97.8220    37.7510
24  United States       NaN                  NaN  -97.8220    37.7510
25  United States  Virginia                Burke  -77.2866    38.7851
26  United States  Virginia              Fairfax  -77.2630    38.8605
27  United States  Virginia                Burke  -77.2866    38.7851
28  United States  Maryland            Baltimore  -76.6949    39.2820
29  United States  Virginia          Springfield  -77.2333    38.7438
30  United States  Virginia               Lorton  -77.2060    38.6911
31  United States  Virginia                Burke  -77.2866    38.7851
32  United States  Virginia         Falls Church  -77.1425    38.8472
33  United States       NaN                  NaN  -97.8220    37.7510
34  United States  Maryland            Frederick  -77.2758    39.4624
35  United States  Virginia              Fairfax  -77.3375    38.8357

    Question 1  … Question 4            Question 5  \
0          4.0  …          63               Taiwan
1          2.0  …          59                India
2          2.0  …          70                India
3          2.0  …          66          South Korea
4          2.0  …          70                   US
5          2.0  …         NaN                  NaN
6          2.0  …          70             pakistan
7          2.0  …         5.2                  NaN
8          2.0  …          71                china
9          4.0  …          71          Uzbekistan
10         2.0  …          66                  USA
11         4.0  …          66                India
12         2.0  …          69              Romania
13         2.0  …          66                India
14         4.0  …          67                China
15         4.0  …          72           Uzbekistan
16         2.0  …          68                  USA
17         2.0  …          60               India
18         4.0  …          67             Ethiopia
19         4.0  …          67        United States
20         4.0  …          74       United States
```

```
21         2.0  …          65                            India
22         4.0  …          67                              USA
23         4.0  …          75                         Nigeria 
24         4.0  …          67                              USA
25         2.0  …          68                           Taiwan
26         2.0  …          72                            India
27         4.0  …          61                          Nigeria
28         4.0  …          64                            India
29         2.0  …          71                            India
30         4.0  …          56                    United States
31         2.0  …          70                            India
32         2.0  …          70  United States ; Pakistan 
33         4.0  …          73                              USA
34         2.0  …          68                          Vietnam
35         4.0  …          66                       Uzbekistan

                Question 6  Question 7                    Question 8  \
0                economics  2021-12-31                 Apple/MacBook
1                  Finance  2021-05-01                 Apple/MacBook
2     Computer Applications  2022-06-01                 Apple/MacBook
3       Business Management  2021-12-31             Microsoft/Windows
4                       Yes  2021-01-01                 Apple/MacBook
5                       NaN         NaN                           NaN
6                   Finance  2022-07-15             Microsoft/Windows
7                       NaN         NaN                           NaN
8                  Economic  2021-12-15             Microsoft/Windows
9    Economics with Finance          NaN                           NaN
10                      BIT  2022-05-01             Microsoft/Windows
11     Computer Engineering  2021-05-31             Microsoft/Windows
12                  Biology  2021-08-26                 Apple/MacBook
13                      Cse  2021-12-25             Microsoft/Windows
14               Management  2021-12-31             Microsoft/Windows
15   Economics with Finance  2022-05-01             Microsoft/Windows
16                Economics  2022-05-31  Microsoft/WindowsApple/MacBook
17             Engineering   2021-01-12             Microsoft/Windows
18       Accounting and ISOM  2021-12-20             Microsoft/Windows
19               Statistics  2021-12-20                 Apple/MacBook
20        Computer Science   2022-05-01             Microsoft/Windows
21        Computer science   2022-04-16             Microsoft/Windows
22  Mathematics & Statistics  2021-12-18  Microsoft/WindowsApple/MacBook
23                Economics  2021-05-14             Microsoft/Windows
24       Systems Engineering  2022-12-19             Microsoft/Windows
25        Civil engineering  2021-05-30             Microsoft/Windows
26                      ECE  2021-12-11                 Apple/MacBook
27       Quantity Surveying  2023-05-24             Microsoft/Windows
28   Bachelor of Engineering  2021-12-30             Microsoft/Windows
29             B.Tech in CS  2021-12-15                 Apple/MacBook
```

```
30             BS accounting  2021-05-01               Microsoft/Windows
31   Bachelor of engineering  2021-12-21  Microsoft/WindowsApple/MacBook
32  Criminology Law and Soci  2021-12-18               Microsoft/Windows
33                 Economics  2022-05-15               Microsoft/Windows
34                   Finance  2022-12-12               Microsoft/Windows
35                Psychology  2021-12-20               OtherAcer/Windows

                           Question 9                Question 10  \
0   Not Working while attending Mason           Some familiarity
1          Working, but not Full Time               Average user
2                      Yes, Full Time           Some familiarity
3   Not Working while attending Mason               Average user
4                      Yes, Full Time           Some familiarity
5                                 NaN                        NaN
6          Working, but not Full Time               Average user
7                                 NaN                        NaN
8                      Yes, Full Time           Some familiarity
9                                 NaN                        NaN
10  Not Working while attending Mason                Little/none
11  Not Working while attending Mason  Frequent use for projects
12         Working, but not Full Time           Some familiarity
13  Not Working while attending Mason               Average user
14  Not Working while attending Mason               Average user
15  Not Working while attending Mason           Some familiarity
16                     Yes, Full Time               Average user
17  Not Working while attending Mason  Frequent use for projects
18  Not Working while attending Mason                Little/none
19         Working, but not Full Time           Some familiarity
20                     Yes, Full Time           Some familiarity
21  Not Working while attending Mason           Some familiarity
22         Working, but not Full Time           Some familiarity
23  Not Working while attending Mason                Little/none
24                     Yes, Full Time               Average user
25                     Yes, Full Time               Average user
26  Not Working while attending Mason               Average user
27  Not Working while attending Mason               Average user
28  Not Working while attending Mason               Average user
29  Not Working while attending Mason               Average user
30                     Yes, Full Time                Little/none
31  Not Working while attending Mason               Average user
32                     Yes, Full Time               Average user
33                     Yes, Full Time               Average user
34  Not Working while attending Mason           Some familiarity
35                     Yes, Full Time           Some familiarity

                  Question 11                Question 12  \
0                 Little/none           Some familiarity
```

1	Average user	Little/none
2	Little/none	Average user
3	Frequent use for projects	Some familiarity
4	Some familiarity	Average user
5	NaN	NaN
6	Average user	Average user
7	NaN	NaN
8	Frequent use for projects	Average user
9	NaN	NaN
10	Little/none	Some familiarity
11	Average user	Frequent use for projects
12	Average user	Average user
13	Average user	Average user
14	Little/none	Little/none
15	Some familiarity	Some familiarity
16	Some familiarity	Some familiarity
17	Frequent use for projects	Average user
18	Little/none	Average user
19	Fluent/expert	Average user
20	Some familiarity	Some familiarity
21	Little/none	Average user
22	Frequent use for projects	Little/none
23	Little/none	Little/none
24	Average user	Average user
25	Some familiarity	Some familiarity
26	Frequent use for projects	Average user
27	Frequent use for projects	Average user
28	Average user	Average user
29	Average user	Frequent use for projects
30	Some familiarity	Some familiarity
31	Frequent use for projects	Frequent use for projects
32	Some familiarity	Some familiarity
33	Average user	Average user
34	Some familiarity	Some familiarity
35	Little/none	Little/none

	Question 13
0	NaN
1	NaN
2	As part of my work, I have access to a lot of …
3	NaN
4	Learn
5	NaN
6	To deduce meaningful interpretation for Financ…
7	NaN
8	NaN
9	NaN

```
10  To see if it is a field I am interested in pur…
11  To learn and explore cutting edge data technol…
12  Learn to select the appropriate tools to condu…
13                                                NaN
14                                  improve my skills
15  My professional goal is to become a Financial …
16  able to compliment my Computational Social Sci…
17                                  To start a company
18  To be able understand and manipulate data to m…
19  My ultimate goal is to get a job as a sports a…
20  To understand how to properly interpret and di…
21  After graduating data analytics, I want myself…
22  Learning how to appropriately handle large qua…
23  To learn and understand the fundamentals and m…
24  The goal is to learn as many programming/stati…
25     Learning analytics tool and being good at them
26                       I want to be as a data analyst
27  It is very relevant to my PhD research area an…
28  Learn how to handle large amount structured an…
29  To become a good Data Analyst and work for top…
30  Get a better understanding of the granularity …
31                                                NaN
32                       Learn how data analytics works
33                    Advance my career in data science
34                                    Getting a good job
35                             to use the skills at work.

[36 rows x 22 columns]
```

[40]:
```
#from geopy import Nominatim
#def get_region_from_coordinates(latitude, longitude):
#    geolocator = Nominatim(user_agent="geoapiExercises")
#    location = geolocator.reverse(f"{latitude}, {longitude}", exactly_one=True)

#    if location is not None:
#        return location.address
#    else:
#        return "Region not found"
#region=get_region_from_coordinates(-97, 37)
#print(region)

#Tried using Geopy Library to Suggest region based on longitude and Laititude
 ↪readings but ran into some issues.
#Therefore replacing it with couldn't calculate
DataFrame1['Region'].fillna('Calculation Failed',inplace=True)
DataFrame1['City / Town / District'].fillna('Calculation Failed',inplace=True)
#Fixing Some Anamolies:
```

```
DataFrame1['Question 3'].replace('1993', '27', inplace=True)
DataFrame1['Question 3'].replace('2 3', '23', inplace=True)
#These all clearly look like typos
DataFrame1['Question 4'].replace('5.2', '52', inplace=True)
DataFrame1['Question 4'].replace('70', '70', inplace=True)
DataFrame1['Question 4'].replace('61', '61', inplace=True)

#Filling in missing height values by first converting to numeric and then␣
↪calculating mean and then going back to string.
DataFrame1['Question 4'] = pd.to_numeric(DataFrame1['Question 4'])
DataFrame1['Question 4'].fillna(DataFrame1['Question 4'].mean(), inplace=True)
DataFrame1['Question 4'] = DataFrame1['Question 4'].astype(int).astype(str)
DataFrame1.isnull().sum()
```

[40]:
```
Time                      0
Status                    0
Duration (Seconds)        0
IP Address                0
Country                   0
Region                    0
City / Town / District    0
Latitude                  0
Longitude                 0
Question 1                0
Question 2                0
Question 3                0
Question 4                0
Question 5                2
Question 6                2
Question 7                3
Question 8                3
Question 9                3
Question 10               3
Question 11               3
Question 12               3
Question 13               9
dtype: int64
```

[41]:
```
DataFrame1['Question 5'] = DataFrame1['Question 5'].str.capitalize()␣
↪#Capitalizing all strings
DataFrame1['Question 5'] = DataFrame1['Question 5'].replace(['Usa', 'United␣
↪states', 'United states ', 'Us'], 'USA') #Deciding on a Unified spelling for␣
↪USA
DataFrame1['Question 5'] = DataFrame1['Question 5'].replace(['United States ;␣
↪Pakistan '], 'USA ; Pakistan') #Capitalising the instance with both USA and␣
↪Pakistan
```

```
DataFrame1['Question 5'].fillna(DataFrame1['Question 5'].mode()[0],␣
↪inplace=True) #Filling Mode value inplace of empty space
DataFrame1.isnull().sum()
```

[41]:
```
Time                      0
Status                    0
Duration (Seconds)        0
IP Address                0
Country                   0
Region                    0
City / Town / District    0
Latitude                  0
Longitude                 0
Question 1                0
Question 2                0
Question 3                0
Question 4                0
Question 5                0
Question 6                2
Question 7                3
Question 8                3
Question 9                3
Question 10               3
Question 11               3
Question 12               3
Question 13               9
dtype: int64
```

[42]:
```
DataFrame1['Question 6'] = DataFrame1['Question 6'].str.capitalize()␣
↪#Capitalizing and deciding on a single common name for various entities
DataFrame1['Question 6'] = DataFrame1['Question 6'].replace('Yes',␣
↪'Unrecognized  Degree') #Just a self-made assumption
DataFrame1['Question 6'] = DataFrame1['Question 6'].replace('Economic',␣
↪'Economics')
DataFrame1['Question 6'] = DataFrame1['Question 6'].replace(['Cse', 'Computer␣
↪science engineering ', 'Computer engineering'], 'Computer Science and␣
↪Engineering')
DataFrame1['Question 6'] = DataFrame1['Question 6'].replace(['Bachelor of␣
↪engineering ', 'Engineering ','Ece'], 'Bachelor of Engineering')
DataFrame1['Question 6'] = DataFrame1['Question 6'].replace('Criminology law␣
↪and soci', 'Criminology Law and Sociology')
DataFrame1['Question 6'] = DataFrame1['Question 6'].replace(['B.tech in␣
↪cs','Bit'], 'Computer Science')
DataFrame1
```

[42]:

```
               Time      Status  Duration (Seconds)       IP Address  \
0    10/3/20 5:49 PM   Completed               11646   100.36.167.226
1    10/3/20 5:49 PM   Completed                 113    98.169.26.134
2    10/3/20 5:49 PM   Completed                 572     98.169.52.23
3    10/3/20 5:50 PM   Completed                 365   69.255.182.253
4    10/3/20 5:51 PM   Completed                  92   138.88.108.230
5    10/3/20 5:59 PM  Incomplete                 221    108.56.134.80
6    10/3/20 6:23 PM   Completed              115983   108.48.175.220
7    10/3/20 7:10 PM  Incomplete                  40    173.67.199.85
8    10/3/20 7:25 PM   Completed                 137     74.96.114.12
9    10/3/20 7:56 PM  Incomplete                 179    107.77.241.32
10   10/3/20 8:14 PM   Completed                 127   138.88.182.245
11   10/3/20 9:38 PM   Completed                 269     72.83.246.60
12   10/3/20 9:41 PM   Completed                 117    96.241.43.105
13  10/3/20 10:49 PM   Completed                5966    71.191.72.223
14  10/3/20 11:25 PM   Completed                 254    73.134.53.227
15  10/4/20 12:28 AM   Completed                 526     73.86.219.52
16   10/4/20 5:22 AM   Completed                 398   69.138.215.127
17   10/4/20 5:36 AM   Completed              128253    72.196.231.70
18   10/4/20 5:47 AM   Completed                 239     100.15.54.17
19  10/4/20 10:32 AM   Completed                 255   184.170.83.239
20  10/4/20 11:03 AM   Completed                 256   100.36.243.102
21  10/4/20 12:34 PM   Completed                 674    108.56.134.80
22   10/4/20 1:36 PM   Completed                 130    73.163.163.82
23   10/4/20 1:50 PM   Completed                 275    138.88.223.87
24   10/4/20 9:28 PM   Completed                 325    100.36.50.114
25   10/5/20 9:37 AM   Completed                 257    108.51.59.195
26   10/5/20 9:36 AM   Completed                 657    72.83.246.219
27  10/5/20 10:36 AM   Completed                1750   100.36.118.186
28  10/5/20 11:02 AM   Completed                 887    172.58.187.98
29  10/5/20 11:17 AM   Completed                 180   72.196.230.201
30  10/5/20 12:52 PM   Completed                 392   108.18.111.204
31  10/5/20 12:56 PM  Incomplete                 192    71.191.72.223
32   10/5/20 1:52 PM   Completed                 277     184.191.72.8
33   10/5/20 3:34 PM   Completed                 119  141.156.160.243
34   10/5/20 3:41 PM   Completed                 149    71.206.51.121
35   10/5/20 5:58 PM   Completed                 285  129.174.252.234

          Country              Region City / Town / District  Latitude  \
0   United States            Virginia                Fairfax  -77.2891
1   United States            Virginia                Fairfax  -77.2630
2   United States            Virginia         Greenway Downs  -77.2331
3   United States            Virginia            Gainesville  -77.6308
4   United States  Calculation Failed     Calculation Failed  -97.8220
5   United States            Virginia                 Reston  -77.3489
6   United States            Virginia              Arlington  -77.0987
7   United States            Virginia                Ashburn  -77.5390
```

8	United States	Virginia	Woodbridge	-77.3197
9	United States	Texas	Calculation Failed	-96.8217
10	United States	Maryland	Takoma Park	-77.0040
11	United States	Virginia	Fairfax	-77.2630
12	United States	Virginia	Chantilly	-77.4512
13	United States	Virginia	Burke	-77.2866
14	United States	Maryland	Hyattsville	-76.8696
15	United States	Maryland	Baltimore	-76.6355
16	United States	Virginia	Arlington	-77.1130
17	United States	Virginia	Springfield	-77.2333
18	United States	Virginia	Woodbridge	-77.3197
19	United States	Virginia	Rustburg	-79.0985
20	United States	Virginia	Fairfax	-77.3375
21	United States	Virginia	Reston	-77.3489
22	United States	Maryland	Baltimore	-76.5570
23	United States	Calculation Failed	Calculation Failed	-97.8220
24	United States	Calculation Failed	Calculation Failed	-97.8220
25	United States	Virginia	Burke	-77.2866
26	United States	Virginia	Fairfax	-77.2630
27	United States	Virginia	Burke	-77.2866
28	United States	Maryland	Baltimore	-76.6949
29	United States	Virginia	Springfield	-77.2333
30	United States	Virginia	Lorton	-77.2060
31	United States	Virginia	Burke	-77.2866
32	United States	Virginia	Falls Church	-77.1425
33	United States	Calculation Failed	Calculation Failed	-97.8220
34	United States	Maryland	Frederick	-77.2758
35	United States	Virginia	Fairfax	-77.3375

	Longitude	Question 1	…	Question 4	Question 5	\
0	38.8209	4.0	…	63	Taiwan	
1	38.8605	2.0	…	59	India	
2	38.9645	2.0	…	70	India	
3	38.8175	2.0	…	66	South korea	
4	37.7510	2.0	…	70	USA	
5	38.9311	2.0	…	67	India	
6	38.8600	2.0	…	70	Pakistan	
7	39.0180	2.0	…	52	India	
8	38.6780	2.0	…	71	China	
9	32.7787	4.0	…	71	Uzbekistan	
10	38.9827	2.0	…	66	USA	
11	38.8605	4.0	…	66	India	
12	38.9036	2.0	…	69	Romania	
13	38.7851	2.0	…	66	India	
14	38.9544	4.0	…	67	China	
15	39.3555	4.0	…	72	Uzbekistan	
16	38.8747	2.0	…	68	USA	

17	38.7438	2.0	…	60	India
18	38.6780	4.0	…	67	Ethiopia
19	37.2544	4.0	…	67	USA
20	38.8357	4.0	…	74	USA
21	38.9311	2.0	…	65	India
22	39.2784	4.0	…	67	USA
23	37.7510	4.0	…	75	Nigeria
24	37.7510	4.0	…	67	USA
25	38.7851	2.0	…	68	Taiwan
26	38.8605	2.0	…	72	India
27	38.7851	4.0	…	61	Nigeria
28	39.2820	4.0	…	64	India
29	38.7438	2.0	…	71	India
30	38.6911	4.0	…	56	USA
31	38.7851	2.0	…	70	India
32	38.8472	2.0	…	70	United states ; pakistan
33	37.7510	4.0	…	73	USA
34	39.4624	2.0	…	68	Vietnam
35	38.8357	4.0	…	66	Uzbekistan

	Question 6	Question 7 \
0	Economics	2021-12-31
1	Finance	2021-05-01
2	Computer applications	2022-06-01
3	Business management	2021-12-31
4	Unrecognized Degree	2021-01-01
5	NaN	NaN
6	Finance	2022-07-15
7	NaN	NaN
8	Economics	2021-12-15
9	Economics with finance	NaN
10	Computer Science	2022-05-01
11	Computer Science and Engineering	2021-05-31
12	Biology	2021-08-26
13	Computer Science and Engineering	2021-12-25
14	Management	2021-12-31
15	Economics with finance	2022-05-01
16	Economics	2022-05-31
17	Bachelor of Engineering	2021-01-12
18	Accounting and isom	2021-12-20
19	Statistics	2021-12-20
20	Computer science	2022-05-01
21	Computer science	2022-04-16
22	Mathematics & statistics	2021-12-18
23	Economics	2021-05-14
24	Systems engineering	2022-12-19
25	Civil engineering	2021-05-30

```
26           Bachelor of Engineering  2021-12-11
27                Quantity surveying  2023-05-24
28           Bachelor of engineering  2021-12-30
29                  Computer Science  2021-12-15
30                     Bs accounting  2021-05-01
31           Bachelor of engineering  2021-12-21
32    Criminology Law and Sociology  2021-12-18
33                         Economics  2022-05-15
34                           Finance  2022-12-12
35                        Psychology  2021-12-20

                      Question 8                          Question 9  \
0                  Apple/MacBook  Not Working while attending Mason
1                  Apple/MacBook         Working, but not Full Time
2                  Apple/MacBook                     Yes, Full Time
3              Microsoft/Windows  Not Working while attending Mason
4                  Apple/MacBook                     Yes, Full Time
5                            NaN                                NaN
6              Microsoft/Windows         Working, but not Full Time
7                            NaN                                NaN
8              Microsoft/Windows                     Yes, Full Time
9                            NaN                                NaN
10             Microsoft/Windows  Not Working while attending Mason
11             Microsoft/Windows  Not Working while attending Mason
12                 Apple/MacBook         Working, but not Full Time
13             Microsoft/Windows  Not Working while attending Mason
14             Microsoft/Windows  Not Working while attending Mason
15             Microsoft/Windows  Not Working while attending Mason
16  Microsoft/WindowsApple/MacBook                     Yes, Full Time
17             Microsoft/Windows  Not Working while attending Mason
18             Microsoft/Windows  Not Working while attending Mason
19                 Apple/MacBook         Working, but not Full Time
20             Microsoft/Windows                     Yes, Full Time
21             Microsoft/Windows  Not Working while attending Mason
22  Microsoft/WindowsApple/MacBook         Working, but not Full Time
23             Microsoft/Windows  Not Working while attending Mason
24             Microsoft/Windows                     Yes, Full Time
25             Microsoft/Windows                     Yes, Full Time
26                 Apple/MacBook  Not Working while attending Mason
27             Microsoft/Windows  Not Working while attending Mason
28             Microsoft/Windows  Not Working while attending Mason
29                 Apple/MacBook  Not Working while attending Mason
30             Microsoft/Windows                     Yes, Full Time
31  Microsoft/WindowsApple/MacBook  Not Working while attending Mason
32             Microsoft/Windows                     Yes, Full Time
33             Microsoft/Windows                     Yes, Full Time
34             Microsoft/Windows  Not Working while attending Mason
```

```
35             OtherAcer/Windows                    Yes, Full Time

                Question 10                Question 11  \
0          Some familiarity                Little/none
1              Average user               Average user
2          Some familiarity                Little/none
3              Average user  Frequent use for projects
4          Some familiarity           Some familiarity
5                       NaN                        NaN
6              Average user               Average user
7                       NaN                        NaN
8          Some familiarity  Frequent use for projects
9                       NaN                        NaN
10              Little/none                Little/none
11  Frequent use for projects             Average user
12         Some familiarity               Average user
13             Average user               Average user
14             Average user                Little/none
15         Some familiarity           Some familiarity
16             Average user           Some familiarity
17  Frequent use for projects  Frequent use for projects
18              Little/none                Little/none
19         Some familiarity              Fluent/expert
20         Some familiarity           Some familiarity
21         Some familiarity                Little/none
22         Some familiarity  Frequent use for projects
23              Little/none                Little/none
24             Average user               Average user
25             Average user           Some familiarity
26             Average user  Frequent use for projects
27             Average user  Frequent use for projects
28             Average user               Average user
29             Average user               Average user
30              Little/none           Some familiarity
31             Average user  Frequent use for projects
32             Average user           Some familiarity
33             Average user               Average user
34         Some familiarity           Some familiarity
35         Some familiarity                Little/none

                Question 12  \
0          Some familiarity
1               Little/none
2              Average user
3          Some familiarity
4              Average user
5                       NaN
```

```
6              Average user
7                       NaN
8              Average user
9                       NaN
10         Some familiarity
11  Frequent use for projects
12             Average user
13             Average user
14              Little/none
15         Some familiarity
16         Some familiarity
17             Average user
18             Average user
19             Average user
20         Some familiarity
21             Average user
22              Little/none
23              Little/none
24             Average user
25         Some familiarity
26             Average user
27             Average user
28             Average user
29  Frequent use for projects
30         Some familiarity
31  Frequent use for projects
32         Some familiarity
33             Average user
34         Some familiarity
35              Little/none

                                          Question 13
0                                                 NaN
1                                                 NaN
2   As part of my work, I have access to a lot of …
3                                                 NaN
4                                               Learn
5                                                 NaN
6   To deduce meaningful interpretation for Financ…
7                                                 NaN
8                                                 NaN
9                                                 NaN
10  To see if it is a field I am interested in pur…
11  To learn and explore cutting edge data technol…
12  Learn to select the appropriate tools to condu…
13                                                NaN
14                                  improve my skills
```

```
15  My professional goal is to become a Financial …
16  able to compliment my Computational Social Sci…
17                              To start a company
18  To be able understand and manipulate data to m…
19  My ultimate goal is to get a job as a sports a…
20  To understand how to properly interpret and di…
21  After graduating data analytics, I want myself…
22  Learning how to appropriately handle large qua…
23  To learn and understand the fundamentals and m…
24  The goal is to learn as many programming/stati…
25     Learning analytics tool and being good at them
26                        I want to be as a data analyst
27  It is very relevant to my PhD research area an…
28  Learn how to handle large amount structured an…
29  To become a good Data Analyst and work for top…
30  Get a better understanding of the granularity …
31                                                 NaN
32                        Learn how data analytics works
33                     Advance my career in data science
34                                     Getting a good job
35                               to use the skills at work.

[36 rows x 22 columns]
```

[ ]:

```
[48]: DataFrame1['Question 6'].fillna('No Undergad Degree', inplace=True)
#If you're among the many who don't have a bachelor's but want to get a␣
↪graduate degree, it is possible. Some universities will
#take into account your professional experience and certain certificates and/or␣
↪diplomas you may have.
#However, it is important for you to do the proper research. (Google)
DataFrame1['Question 7'].fillna(DataFrame1['Question 7'].mode()[0],␣
↪inplace=True) #Filling it with Mode Values as that is the best estimate we␣
↪can have
DataFrame1.isnull().sum()
DataFrame1
```

```
[48]:                 Time      Status  Duration (Seconds)       IP Address  \
0    10/3/20 5:49 PM   Completed               11646   100.36.167.226
1    10/3/20 5:49 PM   Completed                 113    98.169.26.134
2    10/3/20 5:49 PM   Completed                 572     98.169.52.23
3    10/3/20 5:50 PM   Completed                 365   69.255.182.253
4    10/3/20 5:51 PM   Completed                  92   138.88.108.230
5    10/3/20 5:59 PM  Incomplete                 221    108.56.134.80
6    10/3/20 6:23 PM   Completed              115983   108.48.175.220
7    10/3/20 7:10 PM  Incomplete                  40    173.67.199.85
```

8	10/3/20 7:25 PM	Completed	137	74.96.114.12
9	10/3/20 7:56 PM	Incomplete	179	107.77.241.32
10	10/3/20 8:14 PM	Completed	127	138.88.182.245
11	10/3/20 9:38 PM	Completed	269	72.83.246.60
12	10/3/20 9:41 PM	Completed	117	96.241.43.105
13	10/3/20 10:49 PM	Completed	5966	71.191.72.223
14	10/3/20 11:25 PM	Completed	254	73.134.53.227
15	10/4/20 12:28 AM	Completed	526	73.86.219.52
16	10/4/20 5:22 AM	Completed	398	69.138.215.127
17	10/4/20 5:36 AM	Completed	128253	72.196.231.70
18	10/4/20 5:47 AM	Completed	239	100.15.54.17
19	10/4/20 10:32 AM	Completed	255	184.170.83.239
20	10/4/20 11:03 AM	Completed	256	100.36.243.102
21	10/4/20 12:34 PM	Completed	674	108.56.134.80
22	10/4/20 1:36 PM	Completed	130	73.163.163.82
23	10/4/20 1:50 PM	Completed	275	138.88.223.87
24	10/4/20 9:28 PM	Completed	325	100.36.50.114
25	10/5/20 9:37 AM	Completed	257	108.51.59.195
26	10/5/20 9:36 AM	Completed	657	72.83.246.219
27	10/5/20 10:36 AM	Completed	1750	100.36.118.186
28	10/5/20 11:02 AM	Completed	887	172.58.187.98
29	10/5/20 11:17 AM	Completed	180	72.196.230.201
30	10/5/20 12:52 PM	Completed	392	108.18.111.204
31	10/5/20 12:56 PM	Incomplete	192	71.191.72.223
32	10/5/20 1:52 PM	Completed	277	184.191.72.8
33	10/5/20 3:34 PM	Completed	119	141.156.160.243
34	10/5/20 3:41 PM	Completed	149	71.206.51.121
35	10/5/20 5:58 PM	Completed	285	129.174.252.234

	Country	Region	City / Town / District	Latitude	\
0	United States	Virginia	Fairfax	-77.2891	
1	United States	Virginia	Fairfax	-77.2630	
2	United States	Virginia	Greenway Downs	-77.2331	
3	United States	Virginia	Gainesville	-77.6308	
4	United States	Calculation Failed	Calculation Failed	-97.8220	
5	United States	Virginia	Reston	-77.3489	
6	United States	Virginia	Arlington	-77.0987	
7	United States	Virginia	Ashburn	-77.5390	
8	United States	Virginia	Woodbridge	-77.3197	
9	United States	Texas	Calculation Failed	-96.8217	
10	United States	Maryland	Takoma Park	-77.0040	
11	United States	Virginia	Fairfax	-77.2630	
12	United States	Virginia	Chantilly	-77.4512	
13	United States	Virginia	Burke	-77.2866	
14	United States	Maryland	Hyattsville	-76.8696	
15	United States	Maryland	Baltimore	-76.6355	
16	United States	Virginia	Arlington	-77.1130	

17	United States	Virginia	Springfield	-77.2333
18	United States	Virginia	Woodbridge	-77.3197
19	United States	Virginia	Rustburg	-79.0985
20	United States	Virginia	Fairfax	-77.3375
21	United States	Virginia	Reston	-77.3489
22	United States	Maryland	Baltimore	-76.5570
23	United States	Calculation Failed	Calculation Failed	-97.8220
24	United States	Calculation Failed	Calculation Failed	-97.8220
25	United States	Virginia	Burke	-77.2866
26	United States	Virginia	Fairfax	-77.2630
27	United States	Virginia	Burke	-77.2866
28	United States	Maryland	Baltimore	-76.6949
29	United States	Virginia	Springfield	-77.2333
30	United States	Virginia	Lorton	-77.2060
31	United States	Virginia	Burke	-77.2866
32	United States	Virginia	Falls Church	-77.1425
33	United States	Calculation Failed	Calculation Failed	-97.8220
34	United States	Maryland	Frederick	-77.2758
35	United States	Virginia	Fairfax	-77.3375

	Longitude	Question 1	…	Question 4	Question 5 \
0	38.8209	4.0	…	63	Taiwan
1	38.8605	2.0	…	59	India
2	38.9645	2.0	…	70	India
3	38.8175	2.0	…	66	South korea
4	37.7510	2.0	…	70	USA
5	38.9311	2.0	…	67	India
6	38.8600	2.0	…	70	Pakistan
7	39.0180	2.0	…	52	India
8	38.6780	2.0	…	71	China
9	32.7787	4.0	…	71	Uzbekistan
10	38.9827	2.0	…	66	USA
11	38.8605	4.0	…	66	India
12	38.9036	2.0	…	69	Romania
13	38.7851	2.0	…	66	India
14	38.9544	4.0	…	67	China
15	39.3555	4.0	…	72	Uzbekistan
16	38.8747	2.0	…	68	USA
17	38.7438	2.0	…	60	India
18	38.6780	4.0	…	67	Ethiopia
19	37.2544	4.0	…	67	USA
20	38.8357	4.0	…	74	USA
21	38.9311	2.0	…	65	India
22	39.2784	4.0	…	67	USA
23	37.7510	4.0	…	75	Nigeria
24	37.7510	4.0	…	67	USA
25	38.7851	2.0	…	68	Taiwan

```
26    38.8605         2.0  …         72                        India
27    38.7851         4.0  …         61                      Nigeria
28    39.2820         4.0  …         64                        India
29    38.7438         2.0  …         71                        India
30    38.6911         4.0  …         56                          USA
31    38.7851         2.0  …         70                        India
32    38.8472         2.0  …         70  United states ; pakistan 
33    37.7510         4.0  …         73                          USA
34    39.4624         2.0  …         68                      Vietnam
35    38.8357         4.0  …         66                   Uzbekistan

                           Question 6  Question 7  \
0                           Economics  2021-12-31
1                            Finance   2021-05-01
2               Computer applications  2022-06-01
3                 Business management  2021-12-31
4                 Unrecognized  Degree  2021-01-01
5                   No Undergad Degree  2021-12-20
6                             Finance  2022-07-15
7                   No Undergad Degree  2021-12-20
8                           Economics  2021-12-15
9              Economics with finance   2021-12-20
10                   Computer Science  2022-05-01
11   Computer Science and Engineering  2021-05-31
12                            Biology  2021-08-26
13   Computer Science and Engineering  2021-12-25
14                         Management  2021-12-31
15             Economics with finance  2022-05-01
16                          Economics  2022-05-31
17            Bachelor of Engineering  2021-01-12
18                Accounting and isom  2021-12-20
19                         Statistics  2021-12-20
20                  Computer science   2022-05-01
21                  Computer science   2022-04-16
22           Mathematics & statistics  2021-12-18
23                          Economics  2021-05-14
24                Systems engineering  2022-12-19
25                  Civil engineering  2021-05-30
26            Bachelor of Engineering  2021-12-11
27                 Quantity surveying  2023-05-24
28            Bachelor of engineering  2021-12-30
29                   Computer Science  2021-12-15
30                      Bs accounting  2021-05-01
31            Bachelor of engineering  2021-12-21
32      Criminology Law and Sociology  2021-12-18
33                          Economics  2022-05-15
34                            Finance  2022-12-12
```

```
35                     Psychology  2021-12-20

                       Question 8                        Question 9  \
0                   Apple/MacBook  Not Working while attending Mason
1                   Apple/MacBook         Working, but not Full Time
2                   Apple/MacBook                    Yes, Full Time
3               Microsoft/Windows  Not Working while attending Mason
4                   Apple/MacBook                    Yes, Full Time
5                             NaN                               NaN
6               Microsoft/Windows         Working, but not Full Time
7                             NaN                               NaN
8               Microsoft/Windows                    Yes, Full Time
9                             NaN                               NaN
10              Microsoft/Windows  Not Working while attending Mason
11              Microsoft/Windows  Not Working while attending Mason
12                  Apple/MacBook         Working, but not Full Time
13              Microsoft/Windows  Not Working while attending Mason
14              Microsoft/Windows  Not Working while attending Mason
15              Microsoft/Windows  Not Working while attending Mason
16  Microsoft/WindowsApple/MacBook                    Yes, Full Time
17              Microsoft/Windows  Not Working while attending Mason
18              Microsoft/Windows  Not Working while attending Mason
19                  Apple/MacBook         Working, but not Full Time
20              Microsoft/Windows                    Yes, Full Time
21              Microsoft/Windows  Not Working while attending Mason
22  Microsoft/WindowsApple/MacBook         Working, but not Full Time
23              Microsoft/Windows  Not Working while attending Mason
24              Microsoft/Windows                    Yes, Full Time
25              Microsoft/Windows                    Yes, Full Time
26                  Apple/MacBook  Not Working while attending Mason
27              Microsoft/Windows  Not Working while attending Mason
28              Microsoft/Windows  Not Working while attending Mason
29                  Apple/MacBook  Not Working while attending Mason
30              Microsoft/Windows                    Yes, Full Time
31  Microsoft/WindowsApple/MacBook  Not Working while attending Mason
32              Microsoft/Windows                    Yes, Full Time
33              Microsoft/Windows                    Yes, Full Time
34              Microsoft/Windows  Not Working while attending Mason
35              OtherAcer/Windows                    Yes, Full Time

               Question 10                Question 11  \
0         Some familiarity                Little/none
1             Average user               Average user
2         Some familiarity                Little/none
3             Average user  Frequent use for projects
4         Some familiarity           Some familiarity
5                      NaN                        NaN
```

```
6             Average user                Average user
7                      NaN                         NaN
8         Some familiarity  Frequent use for projects
9                      NaN                         NaN
10             Little/none                 Little/none
11  Frequent use for projects               Average user
12        Some familiarity                Average user
13            Average user                Average user
14            Average user                 Little/none
15        Some familiarity            Some familiarity
16            Average user            Some familiarity
17  Frequent use for projects  Frequent use for projects
18             Little/none                 Little/none
19        Some familiarity               Fluent/expert
20        Some familiarity            Some familiarity
21        Some familiarity                 Little/none
22        Some familiarity  Frequent use for projects
23             Little/none                 Little/none
24            Average user                Average user
25            Average user            Some familiarity
26            Average user  Frequent use for projects
27            Average user  Frequent use for projects
28            Average user                Average user
29            Average user                Average user
30             Little/none            Some familiarity
31            Average user  Frequent use for projects
32            Average user            Some familiarity
33            Average user                Average user
34        Some familiarity            Some familiarity
35        Some familiarity                 Little/none

                Question 12  \
0         Some familiarity
1              Little/none
2             Average user
3         Some familiarity
4             Average user
5                      NaN
6             Average user
7                      NaN
8             Average user
9                      NaN
10        Some familiarity
11  Frequent use for projects
12            Average user
13            Average user
14             Little/none
```

15	Some familiarity
16	Some familiarity
17	Average user
18	Average user
19	Average user
20	Some familiarity
21	Average user
22	Little/none
23	Little/none
24	Average user
25	Some familiarity
26	Average user
27	Average user
28	Average user
29	Frequent use for projects
30	Some familiarity
31	Frequent use for projects
32	Some familiarity
33	Average user
34	Some familiarity
35	Little/none

	Question 13
0	NaN
1	NaN
2	As part of my work, I have access to a lot of …
3	NaN
4	Learn
5	NaN
6	To deduce meaningful interpretation for Financ…
7	NaN
8	NaN
9	NaN
10	To see if it is a field I am interested in pur…
11	To learn and explore cutting edge data technol…
12	Learn to select the appropriate tools to condu…
13	NaN
14	improve my skills
15	My professional goal is to become a Financial …
16	able to compliment my Computational Social Sci…
17	To start a company
18	To be able understand and manipulate data to m…
19	My ultimate goal is to get a job as a sports a…
20	To understand how to properly interpret and di…
21	After graduating data analytics, I want myself…
22	Learning how to appropriately handle large qua…
23	To learn and understand the fundamentals and m…

```
24  The goal is to learn as many programming/stati…
25     Learning analytics tool and being good at them
26                        I want to be as a data analyst
27  It is very relevant to my PhD research area an…
28  Learn how to handle large amount structured an…
29  To become a good Data Analyst and work for top…
30  Get a better understanding of the granularity …
31                                                NaN
32                         Learn how data analytics works
33                      Advance my career in data science
34                                      Getting a good job
35                             to use the skills at work.

[36 rows x 22 columns]
```

[51]:
```python
DataFrame1['Question 8'] = DataFrame1['Question 8'].replace('Apple/MacBook',␣
↪'Macbook') #Deciding on one spelling
DataFrame1['Question 8'] = DataFrame1['Question 8'].replace('Acer/Windows',␣
↪'Windows')
DataFrame1['Question 8'] = DataFrame1['Question 8'].replace('Microsoft/
↪Windows', 'Windows')
DataFrame1['Question 8'] = DataFrame1['Question 8'].replace('Microsoft/
↪WindowsApple/MacBook', 'Windows and Macbook')
DataFrame1['Question 8'].fillna(DataFrame1['Question 8'].mode()[0],␣
↪inplace=True) #Using Mode as qualitative data does not have a mean or median.
DataFrame1['Question 9'].fillna(DataFrame1['Question 9'].mode()[0],␣
↪inplace=True) #Using Mode as qualitative data does not have a mean or median.
DataFrame1.head()
```

[51]:
```
               Time     Status  Duration (Seconds)       IP Address  \
0  10/3/20 5:49 PM  Completed               11646  100.36.167.226
1  10/3/20 5:49 PM  Completed                 113    98.169.26.134
2  10/3/20 5:49 PM  Completed                 572     98.169.52.23
3  10/3/20 5:50 PM  Completed                 365   69.255.182.253
4  10/3/20 5:51 PM  Completed                  92   138.88.108.230

         Country              Region City / Town / District  Latitude  \
0  United States            Virginia                Fairfax  -77.2891
1  United States            Virginia                Fairfax  -77.2630
2  United States            Virginia         Greenway Downs  -77.2331
3  United States            Virginia            Gainesville  -77.6308
4  United States  Calculation Failed     Calculation Failed  -97.8220

   Longitude  Question 1  … Question 4  Question 5             Question 6  \
0    38.8209         4.0  …         63      Taiwan              Economics
1    38.8605         2.0  …         59       India                Finance
2    38.9645         2.0  …         70       India  Computer applications
```

```
3    38.8175          2.0  …         66  South korea    Business management
4    37.7510          2.0  …         70          USA  Unrecognized  Degree

   Question 7 Question 8                         Question 9       Question 10  \
0  2021-12-31    Macbook  Not Working while attending Mason  Some familiarity
1  2021-05-01    Macbook         Working, but not Full Time      Average user
2  2022-06-01    Macbook                    Yes, Full Time  Some familiarity
3  2021-12-31    Windows  Not Working while attending Mason      Average user
4  2021-01-01    Macbook                    Yes, Full Time  Some familiarity

              Question 11       Question 12  \
0              Little/none  Some familiarity
1             Average user       Little/none
2              Little/none      Average user
3  Frequent use for projects  Some familiarity
4         Some familiarity      Average user

                                         Question 13
0                                                NaN
1                                                NaN
2  As part of my work, I have access to a lot of …
3                                                NaN
4                                              Learn

[5 rows x 22 columns]
```

[56]:
```python
DataFrame1['Question 13'].fillna('No Comments!',inplace=True) #This is forceful i know but there is no other way to interpret this.
DataFrame1.isnull().sum()
```

[56]:
```
Time                    0
Status                  0
Duration (Seconds)      0
IP Address              0
Country                 0
Region                  0
City / Town / District  0
Latitude                0
Longitude               0
Question 1              0
Question 2              0
Question 3              0
Question 4              0
Question 5              0
Question 6              0
Question 7              0
Question 8              0
```

```
Question 9             0
Question 10            0
Question 11            0
Question 12            0
Question 13            0
dtype: int64
```

1.1.1 Exporting CSV File

[57]:
```python
DataFrame1.to_csv('cleaned_580SurveyCleanup.csv', index=False)
```

1.1.2 Q3

[67]:
```python
DataFrame2 = pd.read_csv('cleaned_580SurveyCleanup.csv')
DataFrame2['Question 1'].describe()
```

[67]:
```
count    36.000000
mean      2.888889
std       1.007905
min       2.000000
25%       2.000000
50%       2.000000
75%       4.000000
max       4.000000
Name: Question 1, dtype: float64
```

This is simple section number and its mean shows that students are more likely or tend to go to section 2.

[68]:
```python
DataFrame2['Question 2'].describe()
```

[68]:
```
count       36
unique       4
top       Male
freq        18
Name: Question 2, dtype: object
```

We got to know that mode value here is Male

[69]:
```python
DataFrame2['Question 3'].astype(float).describe()
```

[69]:
```
count    36.000000
mean     27.055556
std       6.484903
min      22.000000
25%      23.000000
50%      25.000000
75%      27.250000
max      55.000000
```

```
Name: Question 3, dtype: float64
```

We got to know that mean age is 27 and youngest enrolled student in masters program is 22 (Which could be an anamolay).

[70]:
```
DataFrame2['Question 4'].describe()
```

[70]:
```
count    36.000000
mean     67.055556
std       4.863535
min      52.000000
25%      66.000000
50%      67.000000
75%      70.000000
max      75.000000
Name: Question 4, dtype: float64
```

No Anamolies spotted and 67 inches mean height is also plausible

[71]:
```
DataFrame2['Question 5'].describe()
```

[71]:
```
count        36
unique       15
top       India
freq         11
Name: Question 5, dtype: object
```

Most Students have cityzenship of India

[72]:
```
DataFrame2['Question 6'].describe()
```

[72]:
```
count            36
unique           25
top       Economics
freq              5
Name: Question 6, dtype: object
```

Most Students have a background in Economics

[73]:
```
DataFrame2['Question 7'].describe()
```

[73]:
```
count             36
unique            24
top       2021-12-20
freq               6
Name: Question 7, dtype: object
```

The Most repeated date of graduation (expected) is 20-12-2021, however this is most likely an anamoly caused by our forceful population of NAN values with mode

[74]:
```
DataFrame2['Question 8'].describe()
```

[74]:
```
count          36
unique          4
top       Windows
freq           24
Name: Question 8, dtype: object
```

Most Students use Windows Based Laptop

[75]:
```
DataFrame2['Question 9'].describe()
```

[75]:
```
count                                36
unique                                3
top       Not Working while attending Mason
freq                                 20
Name: Question 9, dtype: object
```

Most Students are not working while attending Mason Program. And That is more than half of the class.

[76]:
```
DataFrame2['Question 10'].describe()
```

[76]:
```
count               36
unique               4
top       Average user
freq                15
Name: Question 10, dtype: object
```

[77]:
```
DataFrame2['Question 11'].describe()
```

[77]:
```
count              36
unique              5
top       Little/none
freq               11
Name: Question 11, dtype: object
```

[78]:
```
DataFrame2['Question 12'].describe()
```

[78]:
```
count               36
unique               4
top       Average user
freq                15
Name: Question 12, dtype: object
```

Most Students are Average User in Python and SQL (Which are relatively older than R) which is probably why students lack experience in it

[79]:
```
DataFrame2['Question 13'].describe()
```

[79]:
```
count               36
unique              28
```

```
top       No Comments!
freq                 9
Name: Question 13, dtype: object
```

Statistical Summary here is pretty much useless except for the count. We know that 9 people did not fill out the final question, and that is a huge number. Perhaps this is a good measure that surveys should be short and quick to fill and should not contain subjective questions. We can also infer that they have no goals at all as of now with regards to Data Analytics.

```
[ ]:
```